

Homework 23

Please read Example 6.3 of Durrett. This example is about Poisson random variables, a topic that I mentioned briefly at the end of class.

Please also read from Example 6.25 through the end of Section 6.5. (Example 6.26 is also about Poisson distributions.) Look closely at which types of calculations use the histogram correction and which do not.

Please work the following exercises from Chapter 6 of Durrett:

26 (Use the CLT to estimate the probability that a student gets exactly 100 heads.)

Consider first a single student flipping coins. Let N be the number of heads the student flips, which we treat as a random variable with the binomial(200, $1/2$) distribution. We thus have $E[N] = 100$, $\text{Var}(N) = 50$, and $\sigma(N) \approx 7.071$. Since N is the sum of 200 iid random variables, we expect it (by the CLT) to be approximately normally distributed. As in Homework 22, we estimate $P(N = 100)$ using the histogram correction and the normal approximation.

$$P(N = 100) = P(99.5 \leq N \leq 100.5) \approx P(-0.07071 \leq \text{standard normal} \leq 0.07071) = 2P(\text{standard normal} \leq 0.07071) - 1 \approx 2(0.5279) - 1 = 1.0558 - 1 = 0.0558.$$

The probability that this student does not flip exactly 100 heads is approximately $1 - 0.0558 = 0.9442$.

Since the events “student i does not flip exactly 100 heads” are independent, the probability that no student gets exactly 100 heads is approximately $(0.9442)^{30} \approx 0.1786$.

It is quite likely (over 80% chance) that at least one student will flip exactly 100 heads.

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We roll two dice 180 times. The probability of a double six on one roll of the two dice is $1/36$. The number N of double sixes is a random variable with the binomial(180, $1/36$) distribution.

- (a) We approximate $P(N = 5)$ with the normal approximation. It is reasonable (CLT) to expect N to be approximately normally distributed, since N is a sum of 180 iid random variables (indicators for double six on the various rolls). We have $E[N] = 180/36 = 5$, $\text{Var}(N) = 180(1/36)(35/36) = 175/36$, and $\sigma(N) \approx 2.2048$.

We use the same method as above to approximate $P(N = 5)$:

$$P(N = 5) = P(4.5 \leq N \leq 5.5) \approx P(-0.2268 \leq \text{standard normal} \leq 0.2268) = 2P(\text{standard normal} \leq 0.2268) - 1 \approx 2(0.5910) - 1 = 0.1820.$$

- (b) We learned in Chapter 2, that if p is small and n is large, then a binomial(n , p) probability is approximately a Poisson(np) probability. In this example, np is 5.

$$P(N = 5) \approx 5^5/5! e^{-5} \approx 0.1755.$$

- (c) The exact binomial probability is $C_{180,5} (1/36)^5 (35/36)^{175} \approx 0.1780$.

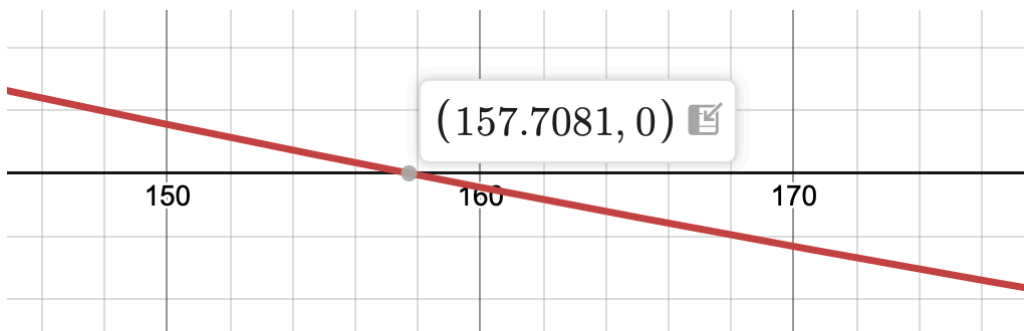
We treat the number N of passengers who show up as a random variable with the binomial($n, 0.85$) distribution, where n is the number of tickets sold. We find the largest value of n such that $P(N \leq 144)$ is at least 0.99.

We have $E[N] = 0.85n$ and $\text{Var}(N) = n(0.85)(0.15)$. The number n will be at least 144, and so N is a sum of a large number (at least 144) of iid random variables. It is thus reasonable to treat N as approximately normally distributed (CLT).

The z-score for 144.5 is $\frac{144.5 - 0.85n}{\sqrt{n(0.85)(0.15)}}$. Looking in the normal table, we see that to get $P(N \leq 144.5) \geq 0.99$, we need the z-score to be at least 2.33. We need to solve the inequality

$$\frac{144.5 - 0.85n}{\sqrt{n(0.85)(0.15)}} \geq 2.33.$$

One way to solve the inequality is by graphing. I graphed $\frac{144.5 - 0.85x}{\sqrt{x(0.85)(0.15)}} - 2.33$ in Desmos. The following screenshot shows a piece of this graph:



The z-scores are thus greater than 2.33 up to $x = 157.7081$. We can take $n = 157$, which is the largest value of n making $P(N \leq 144)$ at least 0.99.

One could also solve this problem using algebra by using the quadratic formula to find x such that

$$\frac{144.5 - 0.85x}{\sqrt{x(0.85)(0.15)}} - 2.33 = 0.$$

Here is the work:

$$\frac{144.5 - 0.85x}{\sqrt{x(0.85)(0.15)}} - 2.33 = 0$$

$$144.5 - 0.85x = 2.33\sqrt{x(0.85)(0.15)}$$

$$(144.5 - 0.85x)^2 = 2.33^2(x(0.85)(0.15))$$

$$144.5^2 - 2(0.85)(144.5)x + 0.85^2x^2 = 2.33^2(x(0.85)(0.15))$$

$$0.85^2x^2 - (2.33^2(0.85)(0.15) + 2(0.85)(144.5))x + 144.5^2 = 0$$

$$0.7225x^2 - 246.34218475x + 20880.25 = 0$$

Therefore, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, where $a = 0.7225$, $b = -246.34218475$, and $c = 20880.25$. The root with the “+” corresponds to z-score -2.33 . The root with the “-” is approximately 157.708, matching what I found earlier with Desmos.

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Let N be the number of seeds in a packet that do not germinate. We treat N as a random variable with the binomial(50, 0.01) distribution. Since this binomial distribution has a large number of trials (N) and a small probability of “success” (0.01), binomial probabilities are approximately Poisson((50)(0.01)) probabilities. The probability of at least 3 non-germinating seeds is $P(N \geq 3) = 1 - P(N \leq 2)$. We approximate $P(N \leq 2)$ as $e^{-0.5} + 0.5e^{-0.5} + \frac{0.5^2}{2}e^{-0.5} = (1.625)e^{-0.5} \approx 0.985612322$, which gives $P(N \geq 3) \approx 0.01438767797$.

Let X be the number of packets out of 4000 that the company has to replace. We treat X as a binomial(4000, 0.01438767797) random variable. Since X is the sum of 4000 iid random variables (indicators for “packet i needs to be replaced”), X is approximately normally distributed (CLT). We have $E[X] = 4000(0.01438767797) \approx 57.55$, $\text{Var}(X) = 4000(0.01438767797)(1 - 0.01438767797) \approx 56.72$, and $\sigma(X) \approx 7.53$.

We use the normal approximation with histogram correction to estimate $P(X > 70) = P(X > 70.5) = 1 - P(X \leq 70.5) \approx 1 - P(\text{standard normal} \leq 1.72) \approx 1 - 0.9573 = 0.0427$.

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Let N be the number of students who enroll in the class. We are given the model $N \sim \text{Poisson}(100)$. We thus have $E[N] = 100$, $\text{Var}(N) = 100$, and $\sigma(N) = 10$. We know from Durrett that a $\text{Poisson}(\lambda)$ distribution with large λ is approximately normally distributed. We approximate $P(N > 120)$ using the normal approximation with histogram correction as follows:

$$P(N > 120) = 1 - P(N \leq 120.5) \approx 1 - P(\text{standard normal} \leq 2.05) \approx 1 - 0.9798 = 0.0202.$$

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Let X_i be the random variable representing the gambler’s winnings on round i . We assume that that X_i are independent random variables. The values of X_i are -1 (probability 0.7), -2 (probability 0.2), and 10 (probability 0.1). We thus have $E[X_i] = (-1)(0.7) + (-2)(0.2) + 10(0.1) = -0.1$ and $E[X_i^2] = (-1)^2(0.7) + (-2)^2(0.2) + 10^2(0.1) = 11.5$. Therefore, $\text{Var}(X_i) = E[X_i^2] - E[X_i]^2 = 11.5 - (-0.1)^2 = 11.49$.

Let $Y = X_1 + \cdots + X_{100}$ be the gambler’s net winnings after 100 rounds. Since Y is the sum of a large number of iid random variables, we may treat it as approximately normally distributed. We have $E[Y] = E[X_1] + \cdots + E[X_{100}] = 100(-0.1) = -10$, $\text{Var}(Y) = \text{Var}(X_1) + \cdots + \text{Var}(X_{100}) = 1149$ (since the X_i are independent), and $\sigma(Y) \approx 33.90$.

We estimate $P(Y \geq 0)$ as follows:

$$P(Y \geq 0) = P(Y > -0.5) = 1 - P(Y \leq -0.5) \approx 1 - P(\text{standard normal} \leq 0.28) \approx 1 - 0.6103 = 0.3897.$$

“A die is rolled repeatedly until the sum of the numbers obtained is larger than 200. What is the probability that you need more than 66 rolls to do this?”

We calculate the probability that in 66 rolls the sum of the numbers is at most 200.

Let $X_1, \dots, X_{66}$ be the numbers rolled in 66 independent tosses of the die, and let $Y = X_1 + \dots + X_{66}$ be the total points rolled in the 66 tosses. From Chapter 1, we know $E[X_i] = 7/2$ and $\text{Var}(X_i) = 105/36$. We have $E[Y] = E[X_1] + \dots + E[X_{66}] = 66 \left(\frac{7}{2}\right) = 231$ and $\text{Var}(Y) = \text{Var}(X_1 + \dots + X_{66}) = \text{Var}(X_1) + \dots + \text{Var}(X_{66}) = 66 \left(\frac{105}{36}\right) = \frac{1155}{6}$, since the X_i are independent random variables.

Since Y is the sum of a large number (66) of iid random variables, it is approximately normally distributed. We approximate $P(Y \leq 200)$ using the normal approximation with histogram correction:

$$P(Y \leq 200) = P(Y < 200.5) = P(\text{standard normal} < -2.20).$$

We calculate $P(\text{standard normal} < -2.20)$ as follows:

$$\begin{aligned} P(\text{standard normal} < -2.20) &= P(\text{standard normal} > 2.20) = 1 - P(\text{standard normal} \leq 2.20) \\ &\approx 1 - 0.9861 = 0.0139. \end{aligned}$$

The probability 0.0139 does not match Durrett’s value, which is 0.0116. Durrett seems to have calculated $P(Y < 200)$, meaning that he would stop rolling if his total at 66 rolls were 200. Here is how to calculate that probability:

$$P(Y < 200) = P(Y < 199.5) = P(\text{standard normal} < -2.27).$$

We calculate $P(\text{standard normal} < -2.27)$ as follows:

$$\begin{aligned} P(\text{standard normal} < -2.27) &= P(\text{standard normal} > 2.27) = 1 - P(\text{standard normal} \leq 2.27) \\ &\approx 1 - 0.9884 = 0.0116. \end{aligned}$$